Week 09: Programming and Research

Programming task

Session 2: deepen this week’s Python.
Predict → change → repair.
Session 3: apply it to the project.

Research task

No new research assignment.
Use the Week 8 proposal and feedback.
Prepare the handoff as project work.

Open learning_materials/Week_09.ipynb. Follow Start Here; keep your work in that notebook. Email: submit.progmis@gmail.com.

Programming task

Session 2 · Python practice before project application.

1. Predict returned values

Do this

Open #session-2: Predict → Run.
Trace `at_least(4/5/6, 5)`.
Predict the pending list.

Observe and explain

Expect False / True / True.
Identify parameters and return.
Explain the filter condition.

2. Change the filter input

Do this

Change only A’s `finished` flag to True.
Predict the new pending list.
Compare `if finished` / `if not finished`.

Observe and explain

Expected pending list: ["C"].
Explain which record changed group.
Try a new minimum with `at_least`.

3. Repair and write assertions

Do this

Run `buggy_at_least` at 4, 5 and 6.
Copy it as `fixed_at_least`.
Repair the boundary.

Observe and explain

Assert below / equal / above.
Keep an already-correct case.
Explain which assertion catches the bug.

Programming practice: done checklist

Keep in your notebook

Prediction before running
Your changed or repaired code
Observed output or error
One explanation of why it happened

Next step

Optional exploration can wait.
Save the notebook and a backup.
Continue to the project application.
Submit to submit.progmis@gmail.com.

Complete the reproduction notes and save the verification evidence.

Project application: reproduce the baseline

Step	What to do / check
Open Session 3	Run the supplied baseline from known inputs.
Record results	Compare the supplied request outcomes and lifecycle totals.
Write your notes	Explain input, processing, output and one limit.
Peer reproduction	Give a peer the run guide; record the result.
Optional	Propose or implement an improvement only if you choose.

Save the reproduced notebook, completed notes and run guide. No new feature is required.